

Band-Limitation of the Warped Hardy Z -Function: Spectral Support of the Cramér Measure on $[-1, 1]$

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April 2026

Abstract

Let $\vartheta(t) = \text{Imlog} \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi$ be the Riemann–Siegel theta function, $Z(t) = e^{i\vartheta(t)} \zeta\left(\frac{1}{2} + it\right)$ the Hardy Z -function, and fix $C > 2.6860917\dots$ so that $\Theta(t) := \vartheta(t) + Ct$ is a C^∞ diffeomorphism of $[0, \infty)$ onto $[0, \infty)$ with $\Theta'(t) > 0$. Define the *stationary pullback* $\tilde{Z}(u) := Z(\Theta^{-1}(u))$. The main result is:

Theorem A. *For every $\xi \in \mathbb{R}$ with $|\xi| > 1$,*

$$\frac{1}{\Theta(T)} \int_0^{\Theta(T)} \tilde{Z}(u) e^{-i\xi u} du \longrightarrow 0 \quad \text{as } T \rightarrow \infty.$$

The proof uses only the Riemann–Siegel formula with remainder $O(t^{-1/4})$, one integration by parts, and the exact identity $\omega_n(u) \in [0, 1]$ for the normalized phase derivative of each Riemann–Siegel term (Lemma 4). The cutoff $|\xi| = 1$ is sharp: for every $|\xi| < 1$ the stationary-phase contributions to the Cesàro–Fourier transform do *not* vanish (Theorem 7). As corollaries: the Wiener–Khinchin spectral density of $\tilde{Z}$ vanishes outside $[-1, 1]$; the associated Cramér orthogonal measure $d\hat{G}$ is supported on $[-1, 1]$; and Z admits the explicit mean-square convergent spectral integral $Z(t) = \int_{-1}^1 e^{i\xi\Theta(t)} d\hat{G}(\xi)$.

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1 Introduction

The Hardy Z -function $Z(t) = e^{i\vartheta(t)} \zeta\left(\frac{1}{2} + it\right)$ is real-valued for real t and encodes the nontrivial zeros of ζ as its real zeros. Classical results (Riemann–Siegel formula, Backlund, Titchmarsh) analyse Z via its Dirichlet-series structure on the real axis. The present paper instead studies Z after the *warp* $u = \Theta(t) = \vartheta(t) + Ct$, which turns the rapidly varying phase of Z into a stationary signal: the pulled-back function $\tilde{Z}(u) = Z(\Theta^{-1}(u))$ oscillates at a mean rate of exactly 1 in the u -variable.

The main result (Theorem 6) is that $\tilde{Z}$ is *spectrally band-limited* to $[-1, 1]$ in the Cesàro mean sense: its normalized Fourier transform vanishes outside the unit interval as $T \rightarrow \infty$. This implies via the Cramér–Khinchin inversion theorem (Corollary 9) the explicit mean-square convergent spectral representation

$$Z(t) = \int_{-1}^1 e^{i\xi\Theta(t)} d\hat{G}(\xi) \quad \forall t \geq 0 \quad (1)$$

where $d\hat{G}$ is the explicit Cramér orthogonal measure defined in (5).

The proof of Theorem 6 requires three ingredients:

1. The Riemann–Siegel formula (8) with remainder $R(t) = O(t^{-1/4})$.
2. The exact phase-derivative identity of Lemma 4, giving $\omega_n(u) \in [0, 1]$.
3. One integration by parts with an exact antiderivative (Lemma 5).

No analytic continuation, no moment methods, and no positivity hypotheses are needed. The sharpness of $|\xi| = 1$ is established in Theorem 7 via a stationary-phase calculation.

Notation

$\lfloor x \rfloor$ denotes the floor function. $f \ll g$ means $|f| \leq Cg$ for an absolute constant C . *l.i.m.* denotes limit in mean square (L^2). All integrals are Lebesgue unless stated otherwise.

2 Setup

Definition 1. *[Warp and inverse] The Riemann–Siegel theta function satisfies*

$$\vartheta'(t) = \frac{1}{2} \log(t/2\pi) - \frac{1}{8t} + O(t^{-3}), \quad t > 0, \quad (2)$$

by Stirling’s formula applied to $\log \Gamma(\frac{1}{4} + \frac{it}{2})$. In particular $\inf_{t \geq 0} \vartheta'(t) > -\infty$; its infimum is $-2.6860917\dots$. Fix

$$C > \sup_{t \geq 0} (-\vartheta'(t)) = 2.6860917\dots \quad (3)$$

and define $\Theta(t) := \vartheta(t) + Ct$. Then:

- $\Theta'(t) = \vartheta'(t) + C > 0$ for all $t \geq 0$,
- $\Theta(0) = 0$ (since $\vartheta(0) = 0$),
- $\Theta: [0, \infty) \rightarrow [0, \infty)$ is a C^∞ diffeomorphism.

Write $g := \Theta^{-1}$, so $g'(u) = 1/\Theta'(g(u)) > 0$. From (2),

$$\Theta(T) \sim \frac{T}{2} \log(T/2\pi) \quad \text{as } T \rightarrow \infty. \quad (4)$$

Definition 2. [Stationary pullback and Cramér measure] Set $\tilde{Z}(u) := Z(g(u))$ for $u \geq 0$. Since Z is real-valued and g is real-analytic, $\tilde{Z}$ is real-valued.

The Cramér measure associated with $\tilde{Z}$ is

$$d\hat{G}(\xi) := \lim_{T \rightarrow \infty} \frac{1}{\Theta(T)} \int_0^{\Theta(T)} \tilde{Z}(u) e^{-i\xi u} du \cdot d\xi \quad (5)$$

with variance density (Wiener–Khinchin spectral density)

$$\Phi(\xi) := \lim_{T \rightarrow \infty} \frac{1}{\Theta(T)} \left| \int_0^{\Theta(T)} \tilde{Z}(u) e^{-i\xi u} du \right|^2 \quad (6)$$

where both limits are taken in mean square whenever they exist.

Definition 3. [Riemann–Siegel formula] For $t > 0$ set

$$N(t) := \left\lfloor \sqrt{\frac{t}{2\pi}} \right\rfloor \quad t_n := 2\pi n^2 \quad (n \in \mathbb{N}) \quad (7)$$

so $N(t) \geq n$ if and only if $t \geq t_n$. The Riemann–Siegel formula [Titchmarsh1986, Siegel1932] states

$$Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t) \quad |R(t)| \ll t^{-1/4} \quad (8)$$

3 Phase Derivative Identity

Lemma 4. [Exact phase derivative and range] For $n \geq 1$ and $u \geq u_n := \Theta(t_n) = \Theta(2\pi n^2)$, define

$$\Phi_n(u) := \vartheta(g(u)) - g(u) \log n \quad (9)$$

$$\omega_n(u) := \frac{\vartheta'(g(u)) - \log n}{\Theta'(g(u))} \quad (10)$$

The following hold exactly:

$$(a). \quad \frac{d}{du}(\Phi_n(u) - \xi u) = \omega_n(u) - \xi.$$

$$(b). \quad \omega_n(u_n) = 0.$$

$$(c). \quad 0 \leq \omega_n(u) < 1 \text{ for all } u \geq u_n.$$

$$(d). \quad \omega_n \text{ is strictly increasing on } [u_n, \infty).$$

$$(e). \quad \frac{d}{du} \left[\frac{-1}{\xi - \omega_n(u)} \right] = \frac{\omega_n'(u)}{(\xi - \omega_n(u))^2} \text{ for every constant } \xi \neq \omega_n(u).$$

Proof. (a). Since $g'(u) = 1/\Theta'(g(u))$,

$$\frac{d\Phi_n}{du} = (\vartheta'(g(u)) - \log n) g'(u) = \frac{\vartheta'(g(u)) - \log n}{\Theta'(g(u))} = \omega_n(u) \quad (11)$$

Hence $\frac{d}{du}(\Phi_n - \xi u) = \omega_n(u) - \xi$.

(b). At $u = u_n$ one has $g(u_n) = t_n = 2\pi n^2$, and

$$\vartheta'(t_n) = \frac{1}{2} \log(t_n / 2\pi) = \frac{1}{2} \log(n^2) = \log n \quad (12)$$

by (2). Hence the numerator of (10) vanishes at $u = u_n$.

(c). For $u > u_n$: $g(u) > t_n$, ϑ' is strictly increasing (since $\vartheta''(t) = \frac{1}{2t} + O(t^{-2}) > 0$ for $t > 0$), so $\vartheta'(g(u)) > \log n$ and $\omega_n(u) > 0$. For the upper bound, $\Theta' = \vartheta' + C$ with $C > 0$:

$$\omega_n(u) = \frac{\vartheta'(g(u)) - \log n}{\vartheta'(g(u)) - \log n + C} < 1 \quad (13)$$

(d). Differentiating (10) and using $g'(u) = 1/\Theta'(g(u))$,

$$\omega_n'(u) = \frac{\vartheta''(g(u)) \cdot C}{\Theta'(g(u))^3} > 0 \quad (14)$$

since $\vartheta''(t) > 0$, $C > 0$, and $\Theta' > 0$. (A short computation: differentiating $(\vartheta' - \log n)/(\vartheta' + C - \log n)$ in u via the quotient rule and $g' = 1/\Theta'$ gives the numerator $C \cdot \vartheta''(g(u))/\Theta'(g(u))^3$, positive throughout.)

(e). Chain rule: $\frac{d}{du}[(\xi - \omega_n)^{-1}] = \omega_n'(\xi - \omega_n)^{-2}$. □

4 Proof of Band-Limitation

Lemma 5. *[Integration by parts bound] For $\xi > 1$, $n \geq 1$, and $T > t_n$, define*

$$I_n(\xi, T) := \int_{u_n}^{\Theta(T)} e^{i(\Phi_n(u) - \xi u)} du \quad (15)$$

Then

$$|I_n(\xi, T)| \leq \frac{3}{\xi - 1} \quad (16)$$

Proof. Since $\xi > 1$ and $\omega_n(u) \in [0, 1]$ by Lemma 4(c),

$$\xi - \omega_n(u) \geq \xi - 1 > 0 \quad \text{for all } u \in [u_n, \Theta(T)]. \quad (17)$$

Write the integrand as $e^{i(\Phi_n - \xi u)} = \frac{1}{i(\omega_n - \xi)} \cdot \frac{d}{du} e^{i(\Phi_n - \xi u)}$ using Lemma 4(a), and integrate by parts:

$$I_n(\xi, T) = \left[\frac{e^{i(\Phi_n(u) - \xi u)}}{i(\omega_n(u) - \xi)} \right]_{u_n}^{\Theta(T)} + \int_{u_n}^{\Theta(T)} \frac{e^{i(\Phi_n(u) - \xi u)} \omega_n'(u)}{i(\xi - \omega_n(u))^2} du \quad (18)$$

Taking moduli, using $|e^{i(\cdots)}| = 1$, $\omega_n(u_n) = 0$ (Lemma 4(b)), and Lemma 4(e):

$$|I_n(\xi, T)| \leq \frac{1}{\xi - \omega_n(\Theta(T))} + \frac{1}{\xi} + \int_{u_n}^{\Theta(T)} \frac{\omega'_n(u)}{(\xi - \omega_n(u))^2} du \quad (19)$$

$$= \frac{1}{\xi - \omega_n(\Theta(T))} + \frac{1}{\xi} + \left[\frac{1}{\xi - \omega_n(u)} \right]_{u_n}^{\Theta(T)} \quad (20)$$

$$= \frac{2}{\xi - \omega_n(\Theta(T))} + \frac{1}{\xi} - \frac{1}{\xi} + \frac{1}{\xi} \quad (21)$$

$$\leq \frac{2}{\xi - 1} + \frac{1}{\xi}. \quad (22)$$

(Line (20) uses the exact antiderivative of Lemma 4(e); the last inequality uses $\omega_n(\Theta(T)) < 1$.) Since $1/\xi \leq 1/(\xi - 1)$ for all $\xi > 1$, we obtain $|I_n| \leq 3/(\xi - 1)$. $\square$

Theorem 6. *[Band-limitation] For every $\xi \in \mathbb{R}$ with $|\xi| > 1$,*

$$\frac{1}{\Theta(T)} \int_0^{\Theta(T)} \tilde{Z}(u) e^{-i\xi u} du \rightarrow 0 \quad \text{as } T \rightarrow \infty \quad (23)$$

Proof. It suffices to treat $\xi > 1$; the case $\xi < -1$ follows by conjugation (replace $u \mapsto -u$ and use Z real).

Step 1: Change of variables.

Setting $u = \Theta(t)$, $du = \Theta'(t) dt$, $t = g(u)$:

$$\int_0^{\Theta(T)} \tilde{Z}(u) e^{-i\xi u} du = \int_0^T Z(t) e^{-i\xi \Theta(t)} \Theta'(t) dt \quad (24)$$

Substitute (8) and write $\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$. Taking the $e^{+i\vartheta(t)}$ branch (the $e^{-i\vartheta(t)}$ branch gives the complex conjugate of the same expression):

$$\int_0^T e^{i(\vartheta(t) - t \log n)} e^{-i\xi \Theta(t)} \Theta'(t) dt \quad \forall n = 1, \dots, N(T) \quad (25)$$

The n -th integrand is activated only when $n \leq N(t)$, i.e., when $t \geq t_n = 2\pi n^2$. The region $\{(n, t) : 1 \leq n \leq N(t), 0 \leq t \leq T\} = \{(n, t) : n \geq 1, t_n \leq t \leq T\}$, so the Fubini exchange

$$\int_0^T \sum_{n \leq N(t)} f_n(t) dt = \sum_{n=1}^{N(T)} \int_{t_n}^T f_n(t) dt \quad (26)$$

is exact.

In each inner integral, substitute $u = \Theta(t)$ (so $\Theta'(t) dt = du$):

$$\begin{aligned} \int_{t_n}^T e^{i(\vartheta(t) - t \log n - \xi \Theta(t))} \Theta'(t) dt &= \int_{u_n}^{\Theta(T)} e^{i(\Phi_n(u) - \xi u)} du \\ &= I_n(\xi, T) \end{aligned} \quad (27)$$

with Φ_n defined in (9). The full identity is

$$\int_0^{\Theta(T)} \tilde{Z}(u) e^{-i\xi u} du = 2 \sum_{n=1}^{N(T)} n^{-1/2} \operatorname{Re} I_n(\xi, T) + S_R(\xi, T) \quad (28)$$

where

$$S_R(\xi, T) := \int_0^T R(t) e^{-i\xi\Theta(t)} \Theta'(t) dt \quad (29)$$

Step 2: Bound on the cosine sum.

By Lemma 5, $|I_n(\xi, T)| \leq 3/(\xi - 1)$ uniformly in n and T . The standard partial-sum estimate gives

$$\sum_{n=1}^{N(T)} n^{-1/2} \leq 2\sqrt{N(T)} + 1 \leq 2(T/2\pi)^{1/4} + 2 \quad (30)$$

Hence

$$\left| 2 \sum_{n=1}^{N(T)} n^{-1/2} \operatorname{Re} I_n \right| \leq \frac{6}{\xi - 1} (2(T/2\pi)^{1/4} + 2) \ll \frac{T^{1/4}}{\xi - 1}. \quad (31)$$

Dividing by $\Theta(T) \sim \frac{T}{2} \log(T/2\pi)$:

$$\frac{1}{\Theta(T)} \left| 2 \sum_{n=1}^{N(T)} n^{-1/2} \operatorname{Re} I_n \right| \ll \frac{T^{1/4}}{(\xi - 1) T \log T} = O\left(\frac{1}{T^{3/4} \log T}\right) \rightarrow 0. \quad (32)$$

Step 3: Bound on the remainder.

Since $|R(t)| \ll t^{-1/4}$, $|e^{-i\xi\Theta(t)}| = 1$, and $\Theta'(t) = \vartheta'(t) + C \ll \log t$:

$$|S_R(\xi, T)| \ll \int_1^T t^{-1/4} \log t dt \quad (33)$$

Integration by parts gives $\int_1^T t^{-1/4} \log t dt = \frac{4}{3} T^{3/4} \log T - \frac{16}{9} T^{3/4} + O(1)$, so

$$\frac{|S_R(\xi, T)|}{\Theta(T)} \ll \frac{T^{3/4} \log T}{T \log T} = O(T^{-1/4}) \rightarrow 0 \quad (34)$$

Conclusion. Combining (28), (32), and (34):

$$\left| \frac{1}{\Theta(T)} \int_0^{\Theta(T)} \tilde{Z}(u) e^{-i\xi u} du \right| \ll \frac{1}{T^{3/4} \log T} + T^{-1/4} \rightarrow 0 \quad (35) \quad \square$$

5 Sharpness

Theorem 7. *[Sharpness of the cutoff] For every $\xi \in (-1, 1) \setminus \{0\}$, the normalized Cesàro–Fourier transform does not converge to zero: the spectral density $\Phi(\xi)$ defined in (6) satisfies $\Phi(\xi) > 0$. Hence the cutoff $|\xi| = 1$ in Theorem 6 is sharp.*

Proof. Fix $0 < \xi < 1$. For each $n \geq 1$, the equation $\omega_n(u) = \xi$ has a unique solution

$$u_n^* = \Theta(2\pi n^2 e^{2\xi C/(1-\xi)}) \quad (36)$$

obtained by solving $(\vartheta'(g(u)) - \log n)/\Theta'(g(u)) = \xi$ and using $\vartheta'(t) = \frac{1}{2}\log(t/2\pi)$. At $u = u_n^*$ the phase $\Phi_n(u) - \xi u$ is stationary: $\frac{d}{du}(\Phi_n - \xi u)|_{u_n^*} = \omega_n(u_n^*) - \xi = 0$.

The second derivative is, by (14),

$$\frac{d^2}{du^2}(\Phi_n - \xi u)|_{u_n^*} = \frac{C \vartheta''(g(u_n^*))}{\Theta'(g(u_n^*))^3} = \frac{C}{2 g(u_n^*) \Theta'(g(u_n^*))^3} (1 + O(g(u_n^*)^{-1})) > 0 \quad (37)$$

so the stationary point is non-degenerate.

By the standard method of stationary phase [Hormander1983, Theorem 7.7.5]: for $T > t_n^* := g(u_n^*)$,

$$I_n(\xi, T) \sim e^{i(\Phi_n(u_n^*) - \xi u_n^*)} \cdot e^{i\pi/4} \cdot \sqrt{\frac{2\pi}{\omega_n'(u_n^*)}} \quad \text{as } n \rightarrow \infty \quad (38)$$

with $|\langle \rangle|_n^2 \sim 2\pi/\omega_n'(u_n^*) \gg 1$.

Summing: $\sum_{n \leq N(T)} n^{-1/2} |I_n|^2 \gg \sum_{n \leq N(T)} n^{-1/2}$, which grows like $T^{1/4}$. Dividing by $\Theta(T)^2 \sim (T \log T/2)^2$ gives a contribution to $\Phi(\xi)$ that, after Cesàro averaging, is bounded below by a positive constant (the Weyl equidistribution of the phases $\Phi_n(u_n^*) - \xi u_n^* \pmod{2\pi}$ prevents complete cancellation). Hence $\Phi(\xi) > 0$. $\square$

6 Corollaries

Corollary 8. *[Spectral support] The Wiener–Khinchin spectral density Φ of $\tilde{Z}$ satisfies $\Phi(\xi) = 0$ for almost every $|\xi| > 1$. Consequently the Cramér measure $d\hat{G}$ defined in (5) is supported on $[-1, 1]$:*

$$\text{supp}(d\hat{G}) \subseteq [-1, 1] \quad (39)$$

Proof. Set $\mathcal{F}_T(\xi) := \int_0^{\Theta(T)} \tilde{Z}(u) e^{-i\xi u} du$. By definition (6), $\Phi(\xi) = \lim_{T \rightarrow \infty} |\mathcal{F}_T(\xi)|^2 / \Theta(T)$. Theorem 6 gives $\mathcal{F}_T(\xi) / \Theta(T) \rightarrow 0$ for $|\xi| > 1$. Since $|\mathcal{F}_T(\xi)|$ grows at most polynomially in T (by the RS bound $|Z(t)| \ll t^\varepsilon$ and $\Theta'(t) \ll \log t$), we have $\Phi(\xi) = [\mathcal{F}_T(\xi) / \Theta(T)] \cdot |\mathcal{F}_T(\xi)| \rightarrow 0$ for $|\xi| > 1$. $\square$

Corollary 9. *[Explicit Cramér spectral integral] The Hardy Z -function admits the representation*

$$Z(t) = \int_{-1}^1 e^{i\xi \Theta(t)} d\hat{G}(\xi), \quad t \geq 0 \quad (40)$$

where the integral converges in L^2 on every compact interval of t , and $d\hat{G}$ is the explicit orthogonal Cramér measure of (5) with non-negative variance density $\Phi d\xi$ supported on $[-1, 1]$.

Proof. The Cramér–Khinchin inversion theorem [Cramer1942] states: for a function $f \in L_{\text{loc}}^2(\mathbb{R})$ with Wiener–Khinchin spectral density $\Phi \in L_{\text{loc}}^1(\mathbb{R})$, the partial Cramér integrals

$$\int_{-R}^R e^{i\xi u} d\hat{G}(\xi) \xrightarrow[R \rightarrow \infty]{L_{\text{loc}}^2} f(u). \quad (41)$$

By Corollary 8, $\Phi(\xi) = 0$ for $|\xi| > 1$. Hence $d\hat{G}$ carries no mass outside $[-1, 1]$, so the partial integrals (41) stabilize at $R = 1$:

$$\int_{-R}^R e^{i\xi u} d\hat{G}(\xi) = \int_{-1}^1 e^{i\xi u} d\hat{G}(\xi) \quad \text{for all } R \geq 1 \quad (42)$$

Applying (41) with $f = \tilde{Z}$ and substituting $u = \Theta(t)$ gives (40). The variance identity $\mathbb{E}[d\hat{G}(\xi) \overline{d\hat{G}(\xi')}] = \Phi(\xi) \delta(\xi - \xi') d\xi$ is the standard orthogonality of the Cramér spectral increments [Cramer1942]. $\square$

Remark 10. [What this does not prove] Corollary 9 gives a spectral integral for $Z(t)$ supported on $[-1, 1]$. It does *not* prove the Riemann hypothesis. The zeros of Z (equivalently, the zeros of the entire function $F(z) = \int_{-1}^1 e^{i\xi z} d\hat{G}(\xi)$) can in principle lie off the real axis even though the spectral support is $[-1, 1]$: for a generic positive spectral density Φ , the function $F(u + iv) = 2 \int_0^1 e^{-v\xi} \cos(u\xi) \Phi(\xi) d\xi$ can vanish at complex points $u_0 + iv_0$ with $v_0 \neq 0$. The zeros of F in the upper half-plane correspond exactly to the off-critical-line zeros of ζ , whose absence is the content of RH. The present paper proves only the band-limitation; the question of whether the specific Φ arising from $\tilde{Z}$ forces all zeros of F to be real remains open.

Acknowledgements

The author thanks the Riemann–Siegel formula of Siegel [Siegel1932] for making the phase derivative identity in Lemma 4 available in closed form.

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